

Human Biology

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Human Biology

- Human biology is an interdisciplinary area of academic study that examines humans through the influences and interplay of many diverse fields such as genetics, evolution, physiology, anatomy, epidemiology, anthropology, ecology, nutrition, population genetics, and sociocultural influences.
- Explanation: Human biology is the study of humans from many different angles. It looks at how our evolution and body systems work, as well as how our environment, diet, and cultures affect us. It also considers how our health changes in different populations and over time. Human biology takes ideas and knowledge from many different areas of science to help us understand how we, as humans, develop, live, and relate to the world around us. It's like using a bunch of different tools to get a complete picture of what makes us who we are and how we fit into our environment.

❖ Most aspects of human biology are identical or very similar to general mammalian biology. In particular, and as examples, humans :

- 1) maintain their body temperature.
- 2) have an internal skeleton.
- 3) have a circulatory system.
- 4) have a nervous system to provide sensory information and operate and coordinate muscular activity.
- 5) have a reproductive system in which they bear live young and produce milk.
- 6) have an endocrine system and produce and eliminate hormones and other bio-chemical signaling agents.
- 7) have a respiratory system where air is inhaled into lungs and oxygen is used to produce energy.
- 8) have an immune system to protect against disease
- 9) Excrete waste as urine and feces.

Vital Parameters

- ❖ Vital signs are measurements of the body's most basic functions, such as breathing rate, blood pressure, pulse rate, and body temperature. These parameters can be measured in a medical setting, at home, or at the site of an emergency. Here's some information about vital signs:
- Blood pressure: The force of blood pushing against artery walls. Normal blood pressure for adults is 120/80 mmHg.
- Heart rate: Also known as pulse, this measures how fast your heart beats. A normal resting heart rate for adults is 60 to 100 beats per minute.
- Respiratory rate: Also known as breathing rate, this measures how many breaths you take per minute. A normal resting respiratory rate for adults is 12 to 20 breaths per minute.
- Body temperature: A normal body temperature for adults is 97.8°F to 99.1°F (36.5°C to 37.3°C). A body temperature higher than normal is called a fever. Vital signs can vary by age, sex, weight, exercise, and overall health. For example, normal heart rates for children vary by age group.

Adult Vital Signs

Normal vital signs vary based on your age, BMI, sex and overall health.

Vital Sign	Adults
Temperature	97.8 F to 99.1 F (36.5 C to 37.3 C).
Blood pressure	90/60 mm Hg to 120/80 mm Hg.
Pulse	60 to 100 beats per minute.
Respiratory rate	12 to 18 breaths per minute.

What Is Pulse Rate

❖ The pulse rate is a measurement of the heart rate, or the number of times the heart beats per minute. As the heart pushes blood through the arteries, the arteries expand and contract with the flow of the blood. Taking a pulse not only measures the heart rate, but also can indicate the following:

- 1) Heart Rhythm
- 2) Strength of the pulse

How to check your pulse rate

❖ As the heart forces blood through the arteries, you feel the beats by firmly pressing on the arteries, which are located close to the surface of the skin at certain points of the body. The pulse can be found on the side of the neck, on the inside of the elbow, or at the wrist. For most people, it is easiest to take the pulse at the wrist. If you use the lower neck, be sure not to press too hard, and never press on the pulses on both sides of the lower neck at the same time to prevent blocking blood flow to the brain. When taking your pulse:

- 1) Using the first and second fingertips, press firmly but gently on the arteries until you feel a pulse.
- 2) Begin counting the pulse when the clock's second hand is on the 12. Count your pulse for 60 seconds (or for 15 seconds and then multiply by four to calculate beats per minute).
- 3) When counting, do not watch the clock continuously, but concentrate on the beats of the pulse.
- 4) If unsure about your results, ask another person to count for you.

What is Body Temperature

- Body temperature is a measure of how hot or cold your body is. It's a crucial aspect of your overall health and helps ensure that your body's processes function correctly. The normal body temperature of a person varies depending on gender, recent activity, food and fluid consumption, time of day, and, in women, the stage of the menstrual cycle. Normal body temperature can range from 97.8 degrees F (or Fahrenheit, equivalent to 36.5 degrees C, or Celsius) to 99 degrees F (37.2 degrees C) for a healthy adult.

Measurement Of Body Temperature

- ❖ A person's body temperature can be taken in any of the following ways:
 - Orally. Temperature can be taken by mouth using either the classic glass thermometer, or the more modern digital thermometers that use an electronic probe to measure body temperature.
 - Rectally. Temperatures taken rectally (using a glass or digital thermometer) tend to be 0.5 to 0.7 degrees F higher than when taken by mouth.
 - Axillary. Temperatures can be taken under the arm using a glass or digital thermometer. Temperatures taken by this route tend to be 0.3 to 0.4 degrees F lower than those temperatures taken by mouth.
 - By ear. A special thermometer can quickly measure the temperature of the ear drum, which reflects the body's core temperature (the temperature of the internal organs).
 - By skin. A special thermometer can quickly measure the temperature of the skin on the forehead.

What is the respiration rate

- The respiration rate is the number of breaths a person takes per minute. The rate is usually measured when a person is at rest and simply involves counting the number of breaths for one minute by counting how many times the chest rises. Respiration rates may increase with fever, illness, and other medical conditions. When checking respiration, it is important to also note whether a person has any difficulty breathing. Normal respiration rates for an adult person at rest range from 12 to 16 breaths per minute.

What is blood pressure

- Blood pressure is the force of the blood pushing against the artery walls during contraction and relaxation of the heart. Each time the heart beats, it pumps blood into the arteries, resulting in the highest blood pressure as the heart contracts. When the heart relaxes, the blood pressure falls. Two numbers are recorded when measuring blood pressure. The higher number, or systolic pressure, refers to the pressure inside the artery when the heart contracts and pumps blood through the body. The lower number, or diastolic pressure, refers to the pressure inside the artery when the heart is at rest and is filling with blood. Both the systolic and diastolic pressures are recorded as "mm Hg" (millimeters of mercury). This recording represents how high the mercury column in an old-fashioned manual blood pressure device (called a mercury manometer or sphygmomanometer) is raised by the pressure of the blood. Today, your doctor's office is more likely to use a simple dial for this measurement"

Types of blood pressure

- Blood pressure is categorized as normal, elevated, or stage 1 or stage 2 high blood pressure:
 - 1) Normal blood pressure is systolic of less than 120 and diastolic of less than 80 (120/80)
 - 2) Elevated blood pressure is systolic of 120 to 129 and diastolic less than 80
 - 3) Stage 1 high blood pressure is systolic is 130 to 139 or diastolic between 80 to 89
 - 4) Stage 2 high blood pressure is when systolic is 140 or higher or the diastolic is 90 or higher"

Thank You